



CLAIMS SET (FOR PROVISIONAL SUPPORT → UTILITY READY)

(You can include these in your provisional — they strengthen priority and later conversion)



INDEPENDENT CLAIM 1 (SYSTEM — BROADEST)

A system for detecting recent substance use, comprising:

- a biological sampling component configured to collect oral fluid from a subject;
 - at least one detection component configured to identify one or more analytes within the collected sample;
 - a processing component configured to generate measurement data corresponding to the detected analytes;
 - a temporal classification engine configured to determine whether the subject is within a defined recent-use temporal window based on the measurement data; and
 - a decision component configured to generate an output classification and initiate one or more system-level actions based on the determined temporal window.
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INDEPENDENT CLAIM 2 (METHOD)

A method for determining recent substance use, comprising:

- collecting an oral fluid sample from a subject;
 - detecting one or more analytes within the sample;
 - generating measurement data corresponding to the detected analytes;
 - determining, using a temporal classification model, whether the subject falls within a defined recent-use window; and
 - generating a classification output and triggering at least one system response based on the determination.
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INDEPENDENT CLAIM 3 (DEVICE)

A portable detection device comprising:

- a sample collection interface;

- a multi-zone detection system configured to detect one or more analytes;
 - a processing unit configured to analyze detection signals; and
 - an output interface configured to provide a classification indicating likelihood of recent substance use within a defined temporal window.
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DEPENDENT CLAIMS (LAYERED PROTECTION)

◆ **Tiered Swab Claims**

4. The system of claim 1, wherein the detection component includes a lateral flow assay having multiple detection zones corresponding to different analyte concentration thresholds.
 5. The system of claim 4, wherein the thresholds include approximately 2 ng/mL, 5 ng/mL, and 10 ng/mL.
 6. The system of claim 4, wherein the detection zones produce visually distinguishable signals.
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◆ **Multi-Channel Device Claims**

7. The system of claim 1, wherein the detection component includes a multi-channel detection system configured to detect both a parent compound and one or more metabolites.
 8. The system of claim 7, wherein the processing component determines a ratio between the parent compound and the one or more metabolites.
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◆ **Temporal Modeling Claims**

9. The system of claim 1, wherein the temporal classification engine uses a ratio-based model.
 10. The system of claim 1, wherein the temporal classification engine uses a probabilistic model.
 11. The system of claim 1, wherein the defined temporal window is approximately two hours.
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◆ Decision System Claims

12. The system of claim 1, wherein the decision component classifies the subject into one of:
- no recent use
 - possible recent use
 - likely recent use
 - high probability recent use
 - inconclusive
13. The system of claim 1, wherein the system initiates an alert or enforcement action based on the classification.
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◆ Network Integration Claims

14. The system of claim 1, wherein the output classification is transmitted to a remote system.
15. The system of claim 14, wherein the remote system includes a law enforcement platform.
16. The system of claim 14, wherein the remote system includes a public health monitoring system.
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◆ Anti-Evasion Claims (VERY IMPORTANT)

17. The system of claim 1, wherein the detection component includes any biological detection method capable of identifying one or more analytes.
18. The system of claim 1, wherein the temporal classification engine includes any algorithm configured to estimate recency of substance use.
19. The system of claim 1, wherein the system operates using hardware, software, or a combination thereof.
20. The system of claim 1, wherein the analyte includes cannabis, alcohol, opioids, stimulants, or any controlled substance.
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FIGURE DESCRIPTIONS (USPTO STYLE)

FIG. 1 — Device Overview

A schematic diagram illustrating an oral fluid collection swab and lateral flow detection device, including sample pad, detection zones, and control region.

FIG. 2 — Tiered Detection Zones

A representation of a multi-zone lateral flow strip showing discrete detection regions corresponding to different analyte concentration thresholds.

FIG. 3 — Multi-Channel Analytical Device

A diagram illustrating a device with multiple detection channels, including pathways for parent compound detection and metabolite detection.

FIG. 4 — Microfluidic Layout

A schematic of a microfluidic system showing sample routing through multiple channels for separation and analysis.

FIG. 5 — Temporal Classification Model

A graphical representation of analyte concentration over time, illustrating a defined temporal window associated with recent use.

FIG. 6 — System Architecture

A block diagram showing integration between detection device, processing system, classification engine, and external network systems.

FIG. 7 — Decision Flow

A flowchart illustrating steps from sample collection to classification and system response.

FIG. 8 — Network Integration

A diagram illustrating communication between the system and external platforms, including enforcement and public health systems.